

BM
4/21/22

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

PRECIPITATION SAMPLES

DATE COLLECTED: 11 APR 2022 Initials: TW

Conversion Factors

10 in Bulk Rain Funnel: NET divided by 51.1 = catch

N/A if Overfilled

12 in Bulk Snow Bucket: NET divided by 69.3 = catch

SITE	TIME (EST)	COLLECTOR VOLUME (ml)	CATCH (mm)	ETI CATCH (mm)	Coll. EFF. (% of ETI)
RG-22	<u>0815</u>	<u>1432</u>	<u>20.7</u>		
RG-1	<u>1305</u>	<u>1552</u>	<u>22.4</u>		
RG-4	<u>1230</u>	<u>1611</u>	<u>23.2</u>		
RG-23	<u>1010</u>	<u>1956</u>	<u>28.2</u>		

Volumes Collected				
	RG-22	RG-1	RG-4	RG-23
Gross g	<u>2392</u>	<u>2512</u>	<u>2573</u>	<u>2917</u>
Tare g	<u>960</u>	<u>960</u>	<u>962</u>	<u>961</u>
Net g	<u>1432</u>	<u>1552</u>	<u>1611</u>	<u>1956</u>

SITE CONTAMINATION

(0 = none, 1 = low -----> 5 = high)

	Fine PM	Coarse PM	Bird FECES	Insects	Color	GB Ring
RG-22	<u>3+</u>	<u>2</u> Bark	<u>0</u>	<u>1</u> beetle	<u>0</u>	<u>2</u>
RG-1	<u>3+</u>	<u>2</u> "	<u>0</u>	<u>3</u> Flying bird water	<u>0</u>	<u>3</u>
RG-4	<u>3+</u>	<u>2</u> y to sect hemlock needles	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>
RG-23	<u>4</u>	<u>3</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>2</u>

dBASE FIELD code? SAMPLE REJECTED @LEVEL = Database CODES checked ?

Precipitation amount
1.13 inches / 28.6 mm
ETI collector

ADDITIONAL COMMENTS: Last bucket collection of season. STA/22 had budburst residue.
Precip was all rain. RG-23 was floating ice cake. Garter snake RG-4

Applicable dBASE FIELD codes:
(See SOP Manual)

970 = Obvious Contamination: Sample REJECTED

969 = Possible Contamination: Check Return Analyses

955 = QNS: Quantity Not Sufficient for analyses (< 100 mL NET catch)

precip-fieldnotes.xls 2017 v RG-4

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

Streamwater Samples

Date Collected:

11 APR 2022

Initials:

TW

SITE	TIME (EST)	TEMP (°C)	GAGE HT.	HYDROGR L/R/F/PK	COLOR (0 = none:1 = low---->5 = high)	SEDIMENT	REMARKS	dBASE FIELD code?
WS-1	1320	7.5°	.301	L	1	0	algae d/s V-NOTCH	
WS-2	1300	6.8°	.280	L	0	0		
WS-3	1245	6.7°	.415	L	1	1	Algae	
WS-4	1400	7.5°	.397	L	0	0		
WS-5	1150	6.7°	.421	L	0	0		
WS-6	1210	7.5°	.328	L	1-	2	Algae	
WS-7	1115	3.1°	.626	L	0	0	Bank snow in stream. 50% SC watershed	
WS-8	1100	2.8°	.515	L	1	1	< 50% snow cover watershed - stream has bank snow	
WS-9	1000	1.1°	.495	L	2+	0	95% SC watershed. stream = 50% snow bridges	
Hubbard BK	1450	5.0°	-	L	1+	0		
Mlake Outlet	0800	3.6°	F		1	0	ICE-OUT!! OTB	

ADDITIONAL COMMENTS: Cool, moist conditions, saturated soils. NFA 50-95% snow cover.

FS Snow data: 0-30 cm D, 0-18 cm SWE.

W-9 phytometer in.

W-1 sensors (exo/SUNA) back in weir after recal. Cleaned leaves.

Applicable dBASE FIELD codes: 911 = High Flow Event in Progress; 912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

955 = QNS: Quantity Not Sufficient for analyses

master sample notes 2017

HUBBARD BROOK ECOSYSTEM STUDY

METROHM / 3 STAR LAB ANALYSES

DATE RUN: 13 APR 2022RUN By: TW

Meter CALIBRATION: See Calibration form for Standard Procedure

Conductivity Correction Factor = 1.14^xat 18.0 °C Lab Temp

Samples:	DATE COLLECTED	Metrohm pH	3 Star pH	SpCond @ 25°C	Remarks Codes
Precipitation					
STATION	<u>11 APR 2022</u>	<u>5.57</u>	<u>5.58</u>	<u>4.5</u>	
RG-1	<u>↓</u>	<u>5.63</u>	<u>5.63</u>	<u>4.9</u>	
RG-4	<u>↓</u>	<u>5.57</u>	<u>5.57</u>	<u>4.7</u>	
RG-23	<u>11 APR 2022</u>	<u>5.57</u>	<u>5.56</u>	<u>4.0</u>	

		Metrohm pH	Metrohm ANC	ANC end pt. pH	3 Star pH	SpCond @ 25°C	Remarks Codes
Streamwater							
WS-1	<u>11 APR 2022</u>	<u>5.52</u>	<u>+5.8</u>	<u>5.15</u>	<u>5.59</u>	<u>12.7</u>	
WS-2	<u>↓</u>	<u>5.50</u>	<u>+4.4</u>	<u>5.16</u>	<u>5.54</u>	<u>8.2</u>	
WS-3	<u>↓</u>	<u>5.66</u>	<u>+9.8</u>	<u>5.26</u>	<u>5.63</u>	<u>9.1</u>	
WS-4	<u>↓</u>	<u>5.96</u>	<u>+15.9</u>	<u>5.18</u>	<u>5.96</u>	<u>9.2</u>	
WS-5	<u>↓</u>	<u>5.44</u>	<u>+2.6</u>	<u>5.25</u>	<u>5.43</u>	<u>9.5</u>	
WS-6	<u>↓</u>	<u>5.45</u>	<u>+0.2</u>	<u>5.28</u>	<u>5.45</u>	<u>9.7</u>	
WS-7	<u>↓</u>	<u>5.75</u>	<u>+11.6</u>	<u>5.29</u>	<u>5.76</u>	<u>8.7</u>	
WS-8	<u>↓</u>	<u>5.51</u>	<u>+0.2</u>	<u>5.42</u>	<u>5.52</u>	<u>9.1</u>	
WS-9	<u>↓</u>	<u>4.68</u>	<u>-22.9</u>	<u>4.41</u>	<u>4.72</u>	<u>13.7</u>	
Hubbard BK	<u>↓</u>	<u>6.07</u>	<u>+21.1</u>	<u>5.21</u>	<u>6.09</u>	<u>9.7</u>	
Mlake Outlet	<u>11 APR 2022</u>	<u>6.46</u>	<u>+68.0</u>	<u>5.06</u>	<u>6.51</u>	<u>30.3</u>	

ADDITIONAL COMMENTS:

Applicable codes: 911= High Flow Event in Progress;

955= QNS. Quantity Not Sufficient for analyses(<100 mL NET catch)

912 = Drought Conditions (svrnge from Standing Pool); 969 = possible problems (note)

metrohm/3 star lab analyses 2017 v4

HBEF Stream and M.Lake Routine Sample DIC Datadate: **12APR2022**

Samples collected 04/11/2022

Aqueous STANDARDS

type	DIC	peak ht	adjusted	ATT -X
	set =	mm	mm	
DIW blank	0	13	0	16
	0	13	0	16
	0	14	0	16
100 µM	100	31	18	16
	100	30	17	16
	100	32	18	16
200 µM	200	52	39	16
	200	53	40	16
	200	52	38	16
400 µM	400	98	85	16
	400	96	83	16
	400	97	83	16

Statistics

Values from matrix at M5 -->

slope = 4 723
 slope ± 0 078
 Y-int = 9 3
 Y-int ± 3 7
 R-sq = 0.9972

Laboratory Conditions

Lab Temp = 18 °C
 Barom. Pressure = 766 mm Hg
 Analyst's Initials TW 59%RH
 NaCO3 Standards:
 0 1M Stock Dated 3/10/22
 0 01M Working Solution Dated 3/10/22

Samples	inj 1	inj 2	inj 3	ATT -X	Avg Peak	±SD Peak	Samples	DIC	Calc	NOTES
	1	2	3					µM/L	err±	120 mA current
ML-70	36	35	34	16	35 0	1 0	ML-70	165	5	
	0	0	0	16	0 0	0 0	0	0	0	
	0	0	0	16	0 0	0 0	0	0	0	
	0	0	0	16	0 0	0 0	0	0	0	
	0	0	0	16	0 0	0 0	0	0	0	
W-1	18	20	22	16	20 0	2 0	W-1	94	9	
W-2	13	13	15	16	13 7	1 2	W-2	65	5	
W-3	13	13	13	16	13 0	0 0	W-3	61	0	2 inj
W-4	12	12	13	16	12 3	0 6	W-4	58	3	
W-5	10	10	12	16	10 7	1 2	W-5	50	5	
W-6	10	9	9	16	9 3	0 6	W-6	44	3	
W-7	12	12	13	16	12 3	0 6	W-7	58	3	
W-8	11	11	11	16	11 0	0 0	W-8	52	0	
W-9	10	10	10	16	10 0	0 0	W-9	47	0	
HBk	12	12	13	16	12 3	0 6	HBk	58	3	
				16	0 0	0 0	0			
Lab Air	18	17	18	16	17 7	0 6	Lab Air	83	3	
Outside Air	12	12	12	16	12 0	0 0	Outside Air	57	0	
Gas 1000	24	25	25	16	24 7	0 6	Gas 1000	117	3	std=83 3
Gas 3000	65	69	72	16	68 7	3 5	Gas 3000	324	17	std=246 3